



Decision Analysis and Resolution (DAR)

DAR Overview

Required PA Information

Intent

Makes and records decisions using a recorded process that analyzes alternatives.

Value

Increases the objectivity of decision-making and the probability of selecting the optimal solution.

Additional Required PA Information

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Explanatory PA Information

Practice Summary

Level 1

- DAR 1.1 Define and record the alternatives.
- DAR 1.2 Make and record the decision.

Level 2

- DAR 2.1 Develop, keep updated, and use rules to determine when to follow a recorded process for criteria-based decisions.
- DAR 2.2 Develop criteria for evaluating alternatives.
- DAR 2.3 Identify alternative solutions.
- DAR 2.4 Select evaluation methods.
- DAR 2.5 Evaluate and select solutions using criteria and methods.

Level 3

- DAR 3.1 Develop, keep updated, and use a description of role-based decision authority.

Additional PA Explanatory Information

Decision analysis and resolution activities involve:

- Developing and updating guidelines to decide which decisions should be subject to a structured, criteria-based decision-making process
- Applying a criteria-based decision-making process for selecting from a set of alternatives

Apply criteria-based decision-making processes to technical or nontechnical alternatives that have multiple possible options for resolution. A decision-making process can be used for decisions throughout the organization.

One key element that contributes to the success of effective decision-making is the involvement of affected stakeholders. Stakeholders can include:

- Those impacted by the decision
- Implementers of the decision

Examples of criteria-based decision-making include:

- Trade studies
- Determination of what solutions will be acquired or developed
- Configuration control board change approvals
- Selection of suppliers
- Risk mitigation choices
- Analysis of alternatives
- A make-or-buy decision
- Choices of manufacturing processes or tools
- Selection of locations, premises, and work facilities
- Changes in organizational structure
- Iteration or release planning

For organizational decisions, specify conditions that require criteria-based decision-making. These conditions could include a combination of:

- Impact on the cost of the decision
- Impact on timelines and schedules
- Impact on the quality of the solution
- Impact on other related processes
- Convenience or disruption to the workforce
- Impact on motivation and morale of the workforce

Criteria-based decision-making processes can vary in formality, type of criteria, and methods used:

- Less formal decisions can be analyzed in a shorter time and use fewer criteria, e.g., effectiveness, cost to implement
- More formal decisions can require more effort, and may include:
 - A plan
 - Multiple reviews to develop and approve criteria
 - Simulations
 - Prototypes
 - Piloting

Related Practice Areas

Planning (PLAN)

Risk and Opportunity Management (RSK)

Context Specific

Agile Development

Context Tag:	Agile Development
Context:	Integrate agile techniques and ceremonies with other processes.

The decisions made by agile teams can include:

- Agile tools to use
- Features to include for each Sprint, story, or epic
- Code refactoring
- Customer acceptance criteria

Development

Context Tag:	CMMI-DEV
Context:	Use processes to develop quality products and services to meet the needs of customers and end users.

Efforts in development can use criteria-based decision-making for alternatives that may include:

- Whether to opt for a short-term code fix that would increase technical debt, or a longer-term solution
- Which design approach to pursue
- Whether to build, buy, or reuse software code
- Whether to invest in automated testing, and to what degree, given the upfront investment needed

Suppliers

Context Tag:	CMMI-SPM
Context:	Use processes to identify, select, and manage suppliers and their agreements.

The acquirer has the ultimate responsibility for ensuring that the appropriate decision-making activities are performed. When the supplier is involved in decisions that affect the overall solution, the acquirer engages with the supplier to provide oversight of the decision process to meet business needs.

Use a repeatable, criteria-based decision-making process for:

- Critical decisions that define and guide the acquisition process
- Critical decisions made with the selected supplier

This decision-making process should be consistent with the acquisition strategy. Revisit these criteria when considering changes or technology additions that affect requirements or other critical project parameters. A formal process also helps the acquirer and supplier to communicate decisions.

Level 1

DAR 1.1

Required Practice Information

Practice Statement

Define and record the alternatives.

Value

Reduces potential rework with a clear definition and understanding of the alternatives.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Decisions do not always consider alternatives. It is important to reach a common understanding of the alternatives, their potential impact, and the necessary decision.

Example Activities

Example Activities	Further Explanation
Define the alternatives.	
Involve stakeholders in defining the alternatives.	

Example Work Products

Example Work Products	Further Explanation
Statement of the alternatives	Describe the alternatives and identify the people involved.

DAR 1.2

Required Practice Information

Practice Statement

Make and record the decision.

Value

Provides a clear understanding of rationale and decisions made and avoids constant revisions and rework.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

The rationale behind decisions may be lost or questioned in the future. Recorded decisions are available for future reference to understand and learn from the decisions made and the issues or contexts associated with them.

Example Activities

Example Activities	Further Explanation
Make and record the decision.	

Example Work Products

Example Work Products	Further Explanation
Recorded decisions	May also include: • Alternatives • Rationale • Selection criteria • People involved

Level 2

DAR 2.1

Required Practice Information

Practice Statement

Develop, keep updated, and use rules to determine when to follow a recorded process for criteria-based decisions.

Value

Reduces costs by focusing on the most important decisions.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Not every decision is significant enough to require a criteria-based decision-making process. Evaluate if a decision is significant based on the work, circumstances, and established guidelines. Consider the cost of making the decision versus the impact of the decision.

Conditions when a criteria-based decision-making process may be needed include when there are:

- Significant adverse effects on cost, quality, resources, or schedule
- Legal or contractual obligations
- Requirements resulting in significantly different alternative solutions
- Issues that have medium-to-high-impact risk
- Changes to work products under configuration management
- Impacts on people's morale, motivation, and convenience

Example Activities

Example Activities	Further Explanation
Develop and record rules and guidelines for when to use a process for criteria-based decision-making.	
Follow rules and guidelines for criteria-based decision-making.	
Communicate rules and guidelines to affected stakeholders.	Inform affected stakeholders when a criteria-based decision-making process will be used.

Example Work Products

Example Work Products	Further Explanation
Rules and guidelines for criteria-based decision-making	
List of recorded criteria-based decisions.	

Related Practice Areas

Risk and Opportunity Management (RSK)

Context Specific

Development

Context Tag:	CMMI-DEV
Context:	Use processes to develop quality products and services to meet the needs of customers and end users.

Examples of activities for which you may use a criteria-based decision-making process include:

- Making design implementation decisions when technical performance failure can cause a catastrophic failure, e.g., safety-of-flight item
- Making decisions with the potential to significantly reduce design risk, engineering changes, cycle time, response time, and production costs, e.g., to use different approaches to assess form and fit capability before releasing engineering drawings and production builds
- Developing new or changing existing requirements resulting in significantly different alternative architectures or designs
- Make, buy, or reuse components
- Selecting testing tools and environment
- Determining alternative software coding approaches

Services

Context Tag:	CMMI-SVC
Context:	Use processes to deliver, manage, and improve services to meet customer needs.

Activities that an organization may use a criteria-based decision-making process for include:

- Selecting elements to include in standard service descriptions
- Selecting, terminating, or renewing suppliers
- Selecting personnel training
- Selecting a transition and support approach, e.g., disaster recovery, service levels

Suppliers

Context Tag:	CMMI-SPM
Context:	Use processes to identify, select, and manage suppliers and their agreements.

Examples of when to use a criteria-based decision-making process include:

- Making decisions to trade off cost, schedule, and performance requirements during an acquisition
- Selecting, terminating, or renewing suppliers
- Selecting a testing environment in which to validate the acquired solution
- Selecting an approach for ongoing support, e.g., disaster recovery, service levels

DAR 2.2**Required Practice Information****Practice Statement**

Develop criteria for evaluating alternatives.

Value

Enables consistent selection of optimal solutions.

Additional Required Information

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Explanatory Practice Information**Additional Explanatory Information**

Both numeric and non-numeric criteria can be used with a process for criteria-based decision-making. Decisions based on explicitly defined criteria can remove barriers to stakeholder agreement and acceptance.

Example Activities

Example Activities	Further Explanation
Define the criteria for evaluating alternative solutions.	Can help to establish boundaries for making decisions.
Define, use, and keep updated the range and weighting for evaluation criteria.	May involve: • Developing weighting for relative importance of evaluation criteria • Identifying risks and impacts • Ranking criteria according to the defined range and weighting to reflect the needs, objectives, and priorities of affected stakeholders

Example Work Products

Example Work Products	Further Explanation
Recorded evaluation criteria	May include: • Rationale for the criteria • Criteria ranking and weighting

DAR 2.3

Required Practice Information

Practice Statement

Identify alternative solutions.

Value

Increases the quality of the solution and customer satisfaction.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

The number of alternatives may be limited at first. Through the analysis process, add other alternatives to the list of potential alternatives. Generate and consider multiple alternatives early in the decision-making process. This makes it easier to choose a solution that best meets the criteria, and to understand the potential consequences of that decision.

Example Activities

Example Activities	Further Explanation
Research information about similar internal or external past decisions.	This can help: • Provide a deeper understanding of the problem • Identify alternatives to consider • Uncover barriers to implementation • Identify lessons learned from similar decisions
Identify additional alternatives to consider.	• Use evaluation criteria as a starting point for identifying alternatives. Evaluation criteria identify the priorities of affected stakeholders and the importance of business, performance, technical, logistical, or other challenges. • Combine key characteristics of existing alternative solutions to generate additional, sometimes stronger, alternative solutions.

Example Activities	Further Explanation
	Solicit alternative solutions from affected stakeholders, e.g., through brainstorming sessions, interviews, and working groups.
Record selected alternatives.	

Example Work Products

Example Work Products	Further Explanation
Recorded alternatives	

Context Specific

Suppliers

Context Tag:	CMMI-SPM
Context:	Use processes to identify, select, and manage suppliers and their agreements.

Suppliers may be needed to provide input to the decision analysis process, e.g., technical expertise outside of acquirer's capabilities. The acquirer has the ultimate decision-making responsibility.

DAR 2.4

Required Practice Information

Practice Statement

Select evaluation methods.

Value

Optimizes the cost, schedule, and performance for the decision being made.

Additional Required Information

This section left blank for future content.

Explanatory Practice Information

Additional Explanatory Information

Evaluation methods are used to determine which alternative meets the stated criteria.

Example Activities

Example Activities	Further Explanation
Select evaluation methods.	Evaluation methods may include: Structured weighting matrix

Example Activities	Further Explanation
	• Testing • Modeling and simulation • Studies or benchmarking • Surveys • Prototypes • Demonstrations • Focus groups • Expert judgment

Example Work Products

Example Work Products	Further Explanation
Selected evaluation methods	

DAR 2.5

Required Practice Information

Practice Statement

Evaluate and select solutions using criteria and methods.

Value

Ensures that the optimal solution is selected.

Additional Required Information

This section left blank for future content.

Explanatory Practice Information

Additional Explanatory Information

A variety of methods may be iteratively used for evaluating and making criteria-based decisions.

Example Activities

Example Activities	Further Explanation
Evaluate proposed alternative solutions following the recorded process for criteria-based decisions.	
Record the results of the evaluation.	Record the rationale for adding new alternatives, adding new methods, changing criteria, and the interim evaluation results.

Example Activities	Further Explanation
Assess the risks associated with implementing the recommended solution.	There can be substantial risk when decisions are made with incomplete information.
Record and communicate the results for the recommended solution to affected stakeholders.	It is important to record both why a solution is selected and why other solutions were rejected.

Example Work Products

Example Work Products	Further Explanation
Evaluation results	
Assessed risks	
Recommended solutions	

Related Practice Areas

Risk and Opportunity Management (RSK)

Level 3

DAR 3.1

Required Practice Information

Practice Statement

Develop, keep updated, and use a description of role-based decision authority.

Value

Reduces business risk by ensuring the appropriate levels of authority are making and approving decisions.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Identify the decision review and approval authority. Approval authority is typically determined by risk and financial, legal, or other business factors in an organization. Determine the organizational approach to be used in decision-making including:

- The level of authority
- Stakeholder involvement
- Required reviews
- Review roles, such as reviewers, facilitators, and approvers

In addition, decision processes in different parts of the organization or at different levels of the organization may be analyzed or approved differently.

Periodically, or on an event-driven basis, e.g., organizational re-structuring; review the decision-making processes to confirm its continued appropriateness, in terms of:

- Speed and timeliness
- Effectiveness
- Availability and accuracy of information
- Responsibility for results
- Breadth of decision impact
- Buy-in for the decision
- Competency
- Decision dependencies
- Coordination of work processes
- Legal responsibilities
- Statutory requirements

Identify and evaluate decisions to modify the decision-making process, roles, or techniques.

Example Activities

Example Activities	Further Explanation
Identify, record, keep updated, and communicate the roles and levels of decision-making authority.	This activity may include: • Periodic review of authority and role definitions • Orientation or training people on organizational decision authority

Example Work Products

Example Work Products	Further Explanation
List of roles and decision-making authority and responsibilities	Decision-making authority may involve roles in areas such as: • Legal or regulatory • Contractual • Financial • Personnel management • Security • Safety • Technical Decision-making information is often captured in a matrix.
List of decision authority levels	Elements of the list may include: • Description of the level • Roles involved include reviewer and approver • Escalation procedures • Communication requirements

